

Multiple Choice (10 marks)

1. Which fraction is greater than 2 over 5?
 - (a) 1 over 5
 - (b) 1 over 10
 - (c) 4 over 10
 - (d) 5 over 10
2. What is the least perfect square with 3 different prime factors?
 - (a) 300
 - (b) 9
 - (c) 900
 - (d) 666
3. Change 34 L₁₈ cL to liters.
 - (a) 34.18 L
 - (b) 35.8 L
 - (c) 34.018 L
 - (d) 214 L
4. What is the sum of all positive integer values of n such that n^2 is a factor of 1200?
 - (a) 42
 - (b) 12
 - (c) 36
 - (d) 39
5. What is the smallest whole number that has a remainder of 1 when divided by 4, a remainder of 1 when divided by 3, and a remainder of 2 when divided by 5?
 - (a) 67
 - (b) 43
 - (c) 57
 - (d) 37

True / False (10 marks)

Answer True or False

6. True or False: The expression $(2 + 5)^2 - 42$ simplifies to 16.
7. True or False: All octagons are polygons because they have at least 3 sides.
8. True or False: The average number of square miles per county in Georgia is 274 square miles.
9. True or False: The correct answer to 'When the expression $-2x^2 - 20x - 53$ is written in the form $a(x + d)^2 + e$, where a , d , and e are constants, what is the sum $a + d + e$?' is '0'.
10. True or False: The expression 8×80 equals 720.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze Bob's morning routine of rolling a die and determine the expected number of rolls he will make in a non-leap year. Explain your reasoning and calculations in detail.
12. Discuss the outcomes of flipping a fair coin in two different games: one requiring 45%-55% heads and the other needing more than 80% heads. Which number of flips would optimize your chances of winning for each game?

End of Paper